

MedVeda-AI

System Documentation

1. Introduction

MedVeda-AI is an advanced AI-powered medical assistant designed to deliver accurate, context-aware, and user-friendly healthcare information. In today's digital landscape, users frequently rely on online sources for medical queries, often encountering fragmented, inconsistent, or misleading information.

MedVeda-AI addresses this gap by combining **Large Language Models (LLMs)** with **Retrieval-Augmented Generation (RAG)**. Instead of generating answers purely from model memory, the system retrieves relevant information from trusted medical sources and uses it to construct reliable responses.

The system is designed with a strong focus on:

- **Accuracy**
- **User experience**
- **Safety-aware responses**

It acts as an **intelligent medical information assistant**, not a replacement for professional healthcare.

2. Purpose

The primary goal of MedVeda-AI is to make high-quality medical knowledge accessible and understandable.

Key objectives include:

- **Simplifying Medical Knowledge**
Converting complex terminology into clear, user-friendly language.
- **Providing Instant Information**
Delivering fast and relevant responses to health-related queries.
- **Reducing Misinformation**
Using RAG to ground answers in trusted sources like medical encyclopedias.
- **Improving User Experience**
Ensuring responses are not only accurate but also concise and context-aware.

3. Problem Statement

Current problems in digital healthcare:

1. Information Overload

Users get too many results with no clear answer.

2. AI Hallucination

Models generate incorrect but confident responses.

Solution

MedVeda-AI uses:

Retrieval-Augmented Generation (RAG)

This ensures:

- Answers are **grounded in real data**
- Reduces hallucination
- Improves trust

4. System Overview

MedVeda-AI is a full-stack AI system consisting of frontend, backend, and AI pipeline layers.

A. Frontend (User Interface Layer)

Built using:

- React (Vite)
- Tailwind CSS
- Framer Motion

Features:

- Clean and minimal chat interface
- Persistent chat history

- Responsive design (mobile + desktop)
- Smooth animations for better UX

B. Backend (Application Layer)

Built using:

- FastAPI

Responsibilities:

- API handling
- Authentication (Google OAuth)
- Session management
- Communication between frontend and AI system

AI & Data Layer (Core Intelligence)

This is the most critical component.

Technologies:

- LangChain + LangGraph → workflow orchestration
- Pinecone → vector database
- Groq (Llama 3.3) → LLM inference
- DuckDuckGo → real-time search

How the System Thinks (Core Idea)

Unlike basic chatbots, MedVeda-AI:

1. Understands the **type of question**
2. Retrieves **relevant knowledge**
3. Generates a **context-aware answer**
4. Applies **safety rules only when needed**

5. Functional Requirements

- User authentication (Google OAuth)

- Persistent chat history
- Context-aware conversation (last 10 messages)
- Intelligent source selection (DB vs Web)
- Dynamic response formatting
- Conditional disclaimer system
- Automatic chat title generation

6. Methodology

Development followed a structured pipeline:

1. Data Processing

- Medical PDFs → chunked → embeddings

2. Vector Storage

- Stored in Pinecone for semantic search

3. Workflow Design

- LangGraph used to define AI decision flow

4. Retrieval Logic

- Database first → Web fallback

5. Response Generation

- LLM uses retrieved context

6. Post-processing

- Safety check
- Formatting based on query type

7. Response Optimization

One of the biggest improvements in MedVeda-AI is **response control**.

Problem:

Traditional AI gives:

Too long answers

- Same format for every query
- Unnecessary disclaimers

Solution:

MedVeda-AI uses:

1. Intent-Based Answering

- Fact → short answer
- Explanation → moderate
- Medical → structured

2. Conditional Disclaimer System

Risk Level	Behavior
Low	No disclaimer
Medium	Soft advice
High	Strong warning

3. Human-Friendly Tone

- Simple language
- Direct answers
- No robotic output

8. Future Work

- Mobile application (React Native)
- Voice assistant integration
- Multi-language support

- Expanded medical dataset

9. Conclusion

MedVeda-AI demonstrates how modern AI systems can be designed to balance:

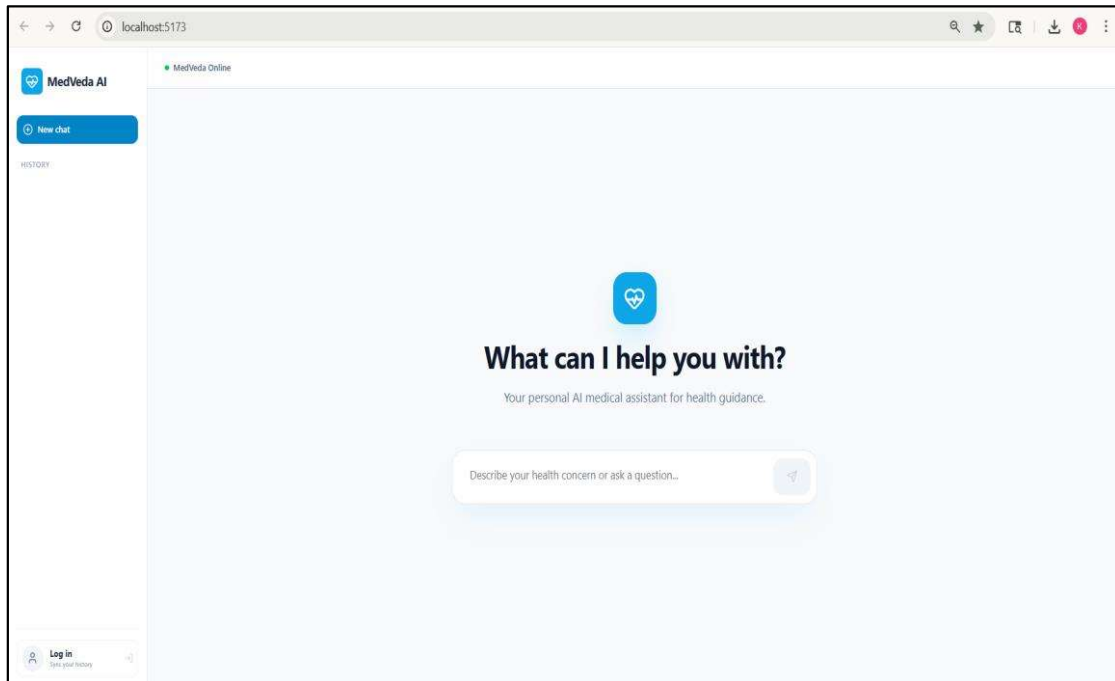
- **Accuracy**
- **Performance**
- **User experience**
- **Safety**

By integrating RAG, intelligent workflows, and adaptive response design, the system moves beyond traditional chatbots and becomes a **real-world AI application**.

10. References

- LangChain Documentation
- FastAPI Documentation
- Groq Platform
- Pinecone Database
- React Documentation
- The Gale Encyclopedia of Medicine
- The Merck Manua

ScreenShort



System Architecture

